

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models
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Github Repository	Click Here

1 Objective

To implement Linear, Ridge, Lasso and Elastic Net regression models to predict the loan sanctioned amount, tune regularization hyperparameters via GridSearchCV with 5-fold cross-validation, and analyze overfitting/underfitting and bias-variance trade-offs.

2 Problem Statement

Given customer, income, credit and property attributes, predict the continuous target **Loan Sanction Amount (USD)** using baseline and regularized linear regression models.

3 Dataset Description

Dataset Name	train.csv (Predict Loan Amount Data)
Dataset Source	Kaggle – Predict Loan Amount Data
Number of Samples	29322 (after dropping rows with missing target)
Number of Features	20 predictors (Customer ID, Name, Property ID dropped as identifiers)
Target Variable	Loan Sanction Amount (USD) – continuous
Missing Values	Present in Gender, Income, Income Stability, Type of Employment, Current Loan Expenses, Dependents, Credit Score, Has Active Credit Card, Property Age, Property Location, Co-Applicant, Property Price – median/most-frequent imputed
Train-Test Split	80% train / 20% test, random_state=42

4 Exploratory Data Analysis

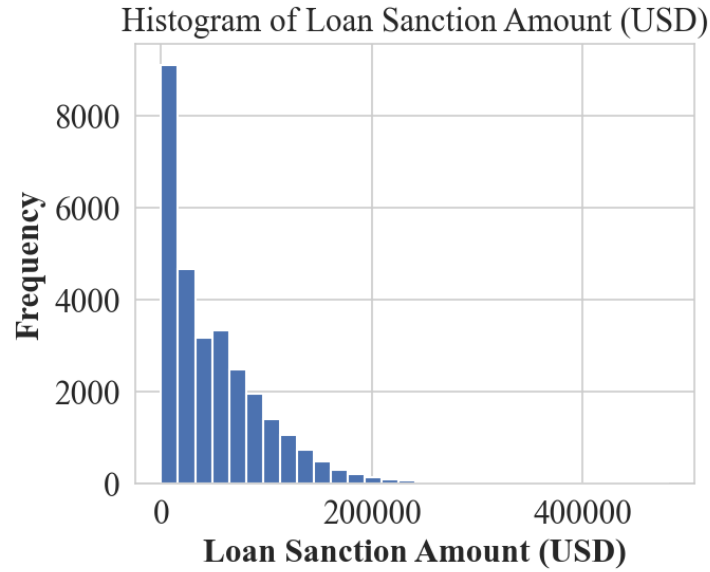


Figure 1: Histogram of Loan Sanction Amount (USD) – target distribution

Inference: The target distribution is right-skewed with a spike at 0 (loans not sanctioned) and a long tail of large sanctioned amounts, which explains the relatively modest R^2 achieved by linear models on this dataset.

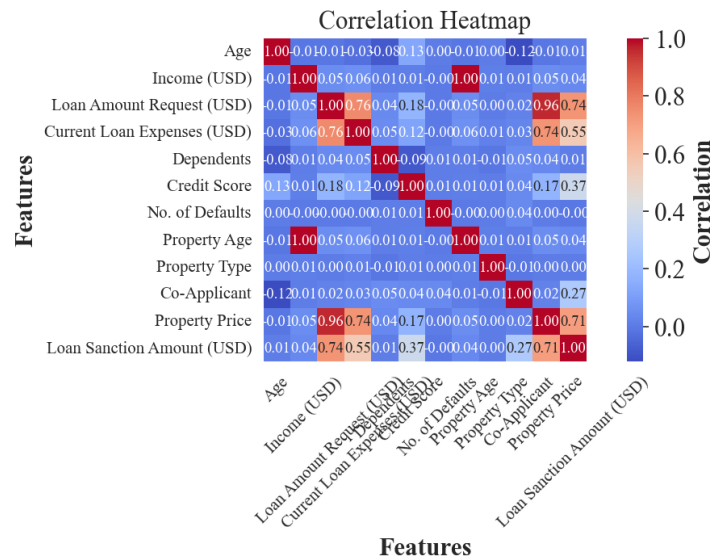


Figure 2: Correlation Heatmap

Inference: Loan Amount Request (USD) and Property Price show the strongest positive correlation with the sanctioned amount, matching their dominance in the feature-selection and coefficient tables.

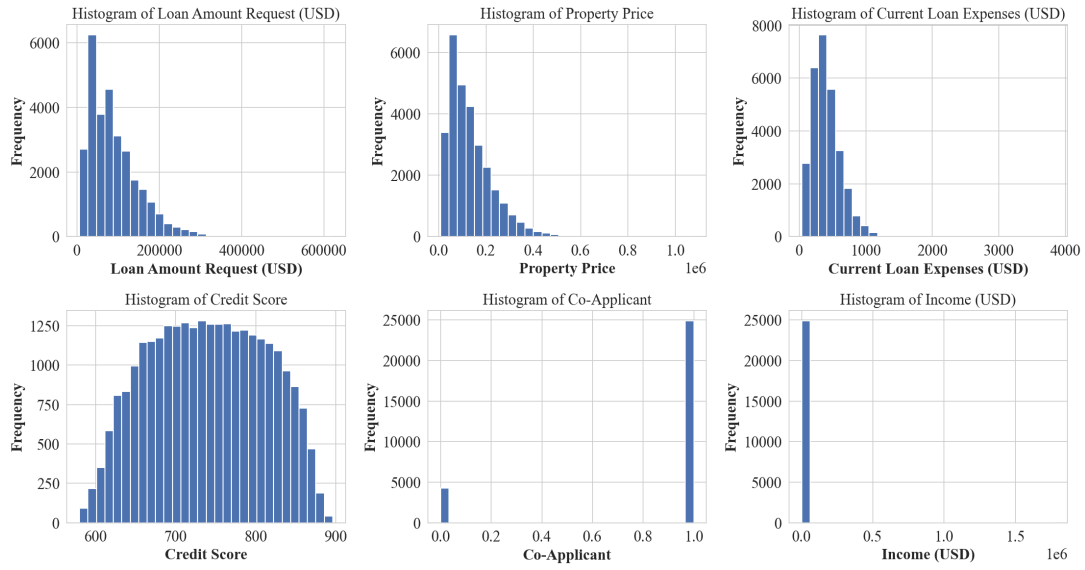


Figure 3: Histogram grid of the top 6 features (by mutual information)

Inference: Income and loan-request features are heavily right-skewed with long tails, while Credit Score is closer to symmetric, indicating that standardization (used in the preprocessing pipeline) is important before fitting linear models.

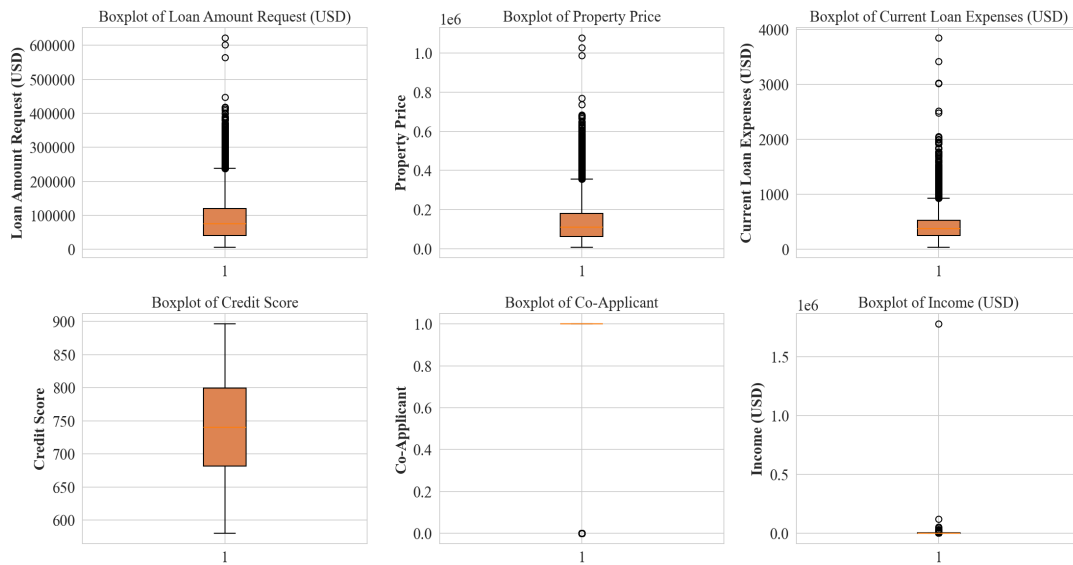


Figure 4: Boxplot grid of the top 6 features

Inference: Income (USD) and Property Price show substantial high-value outliers, which likely contribute disproportionately to squared-error-based metrics like MSE/RMSE.

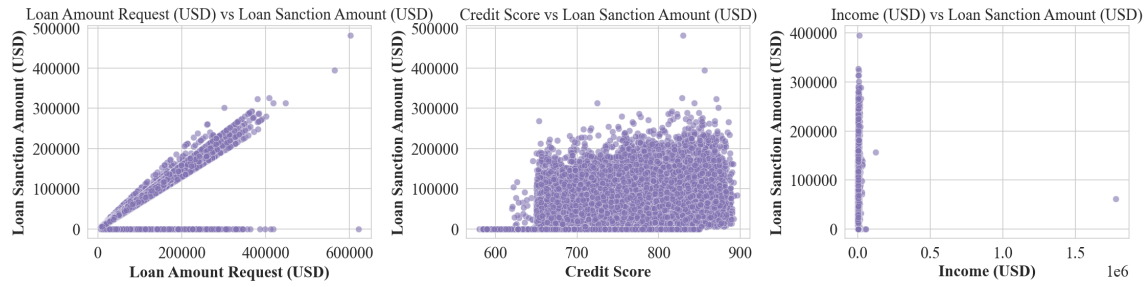


Figure 5: Feature vs. target scatter plots: Loan Amount Request, Credit Score, and Income (USD) vs. Loan Sanction Amount

Inference: Loan Amount Request shows the clearest positive linear trend with the sanctioned amount, Credit Score shows a weaker but visible positive trend, while Income shows a noisy, weakly positive relationship with several high-leverage outliers.

5 Data Preprocessing

- Sentinel value `-999` replaced with `NaN` in Current Loan Expenses, Co-Applicant, Property Price and the target column.
- Rows with a missing target dropped.
- Numeric features: median imputation followed by `StandardScaler`.
- Categorical features: most-frequent imputation followed by `OneHotEncoder(handle_unknown='ignore')`.
- Combined via a `ColumnTransformer` inside a single `Pipeline` to avoid data leakage across CV folds.
- 80/20 train-test split, `random_state=42`.

6 Machine Learning Algorithm

Linear Regression

Models the target as a linear combination of input features; minimizes sum of squared residuals; serves as the baseline.

Ridge Regression

Adds an ℓ_2 penalty $\alpha \sum \beta_i^2$ to the loss, shrinking coefficients and reducing variance without eliminating features.

Lasso Regression

Adds an ℓ_1 penalty $\alpha \sum |\beta_i|$, driving some coefficients exactly to zero and performing implicit feature selection.

Elastic Net

Combines ℓ_1 and ℓ_2 penalties via a mixing ratio (`l1_ratio`), balancing Ridge's stability with Lasso's sparsity.

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Selection

Parameter	Search Space
Ridge α	0.01, 0.1, 1, 10, 100
Lasso α	0.001, 0.01, 0.1, 1, 10
Elastic Net α	0.01, 0.1, 1, 10
Elastic Net <code>l1_ratio</code>	0.2, 0.5, 0.8
CV	GridSearchCV, <code>cv=5</code> , <code>scoring='r2'</code>

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid	$\alpha = 1$	0.6426
Lasso Regression	Grid	$\alpha = 10$	0.6429
Elastic Net Regression	Grid	$\alpha = 0.01$, <code>l1_ratio</code> =0.8	0.6424

9 Results

Table 2: Cross-Validation Performance (K=5, on training set)

Model	MAE	MSE	RMSE	R^2
Linear Regression	18903.17	8.3795e8	28947.41	0.6425
Ridge Regression	18900.06	8.3793e8	28947.08	0.6426
Lasso Regression	18890.38	8.3698e8	28930.56	0.6429
Elastic Net Regression	18898.81	8.3820e8	28951.70	0.6424

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R ²
Linear Regression	18761.75	7.7042e8	27756.40	0.6575
Ridge Regression	18757.79	7.7021e8	27752.66	0.6576
Lasso Regression	18744.35	7.7017e8	27751.91	0.6576
Elastic Net Regression	18762.97	7.7071e8	27761.58	0.6574

Table 4: Coefficient Comparison (Top 10 by |Linear coefficient|)

Feature	Linear	Ridge	Lasso	Elastic Net
Property Age	-45952.81	-30811.19	-17.70	-1917.50
Income (USD)	45931.28	30788.93	-0.00	1891.09
Loan Amount Request (USD)	33139.94	33125.31	33128.84	32300.21
Profession_Unemployed	19082.43	9755.46	0.00	353.34
Profession_Pensioner	13061.20	13507.80	15733.05	11010.97
Co-Applicant	12583.22	12581.98	12562.57	12535.60
Profession_Maternity leave	-11705.63	-5544.78	-0.00	-200.85
Credit Score	10269.52	10267.13	10279.23	10280.53
Profession_State servant	-6583.33	-5929.31	-51.65	-3781.75
Profession_Working	-6578.33	-5919.88	-232.15	-3863.70

10 Result Visualization

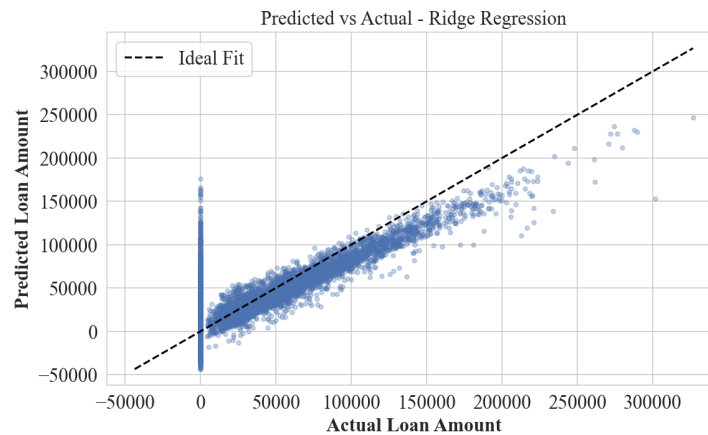


Figure 6: Predicted vs Actual – Ridge Regression

Inference: Predictions follow the ideal-fit line reasonably well for low-to-moderate loan amounts but increasingly underpredict at the high end, showing the linear model struggles to capture the heavy right tail of the target distribution.

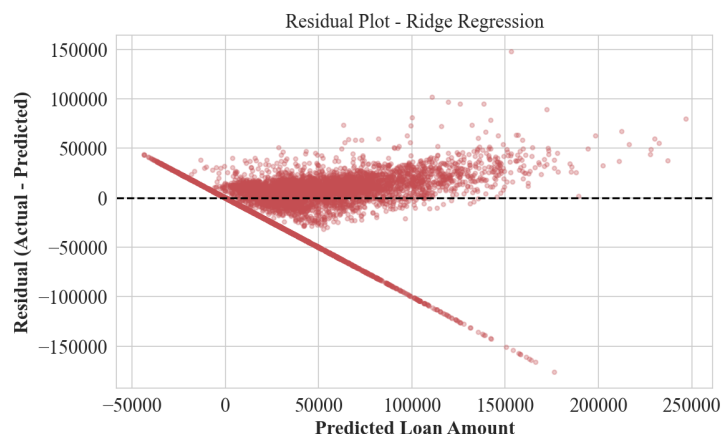


Figure 7: Residual Plot – Ridge Regression

Inference: Residuals fan out and become increasingly negative as predicted values grow, indicating heteroscedasticity and a systematic underprediction bias for high-value loans – a sign the model’s linear assumption is imperfect for this skewed target.

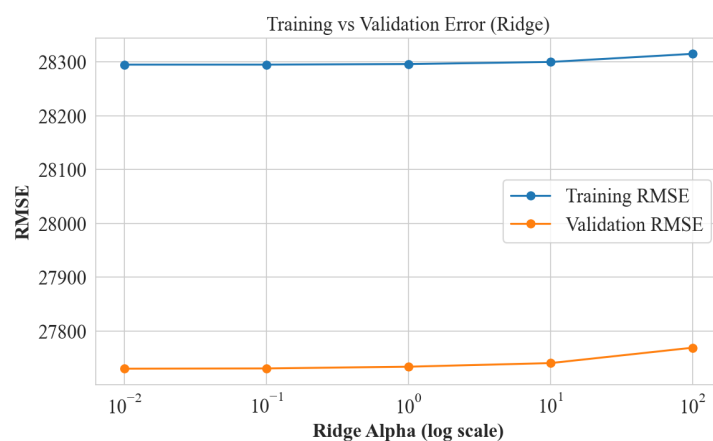


Figure 8: Training vs Validation Error (Ridge)

Inference: Training and validation RMSE remain close together across the alpha range (0.01 to 100), with only a mild increase in both at very high alpha (100), indicating the model is closer to underfitting than overfitting and that regularization strength has limited impact on this dataset.

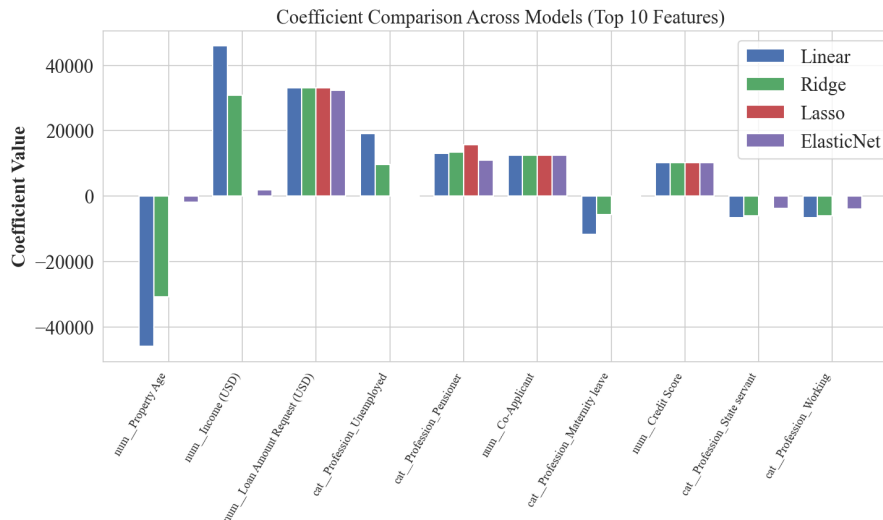


Figure 9: Coefficient Comparison Across Models (Top 10 Features)

Inference: Lasso aggressively shrinks Property Age and Income coefficients toward zero (feature elimination), while Ridge shrinks them more moderately without eliminating them; Elastic Net sits between the two, confirming its role as a compromise between sparsity and stability.

11 Overfitting and Underfitting Analysis

Training and test R^2 values are close (0.643 vs 0.658) across all four models, indicating low variance and no significant overfitting; if anything, the models slightly underfit, since R^2 plateaus around 0.64-0.66 regardless of regularization strength, suggesting the linear model family cannot fully capture the underlying (likely non-linear) relationship between features and sanctioned loan amount. Increasing regularization strength (higher Ridge α) did not meaningfully close this gap, confirming the limiting factor is model bias rather than variance.

12 Bias–Variance Analysis

- **Linear Regression:** Lowest regularization (no shrinkage) but nearly identical R^2 to the regularized variants, suggesting bias dominates over variance for this dataset – the model is not overfitting, so additional regularization yields negligible improvement.
- **Ridge and Elastic Net:** Reduce coefficient magnitude and variance for correlated/redundant features (e.g. Property Age and Income, which are highly collinear) without materially changing predictive R^2 .
- **Lasso:** Produces the sparsest coefficients (several Profession dummies driven to exactly 0), trading a small amount of bias for a simpler, more interpretable model while achieving marginally the best test R^2 (0.6576) among all four.

13 Discussion

All four models converge to nearly identical performance (test $R^2 \approx 0.657$ - 0.658), indicating that regularization strength has limited effect once features are properly standardized/encoded – the bottleneck is model bias (linear form) rather than overfitting. Lasso’s feature sparsification offers an interpretability advantage with no accuracy cost, making it an attractive choice when a simpler model is desired.

14 Observations and Conclusion

Linear, Ridge, Lasso and Elastic Net regression achieved comparable predictive performance ($R^2 \approx 0.66$) on the loan sanction amount prediction task. Lasso marginally outperformed the others on the test set while also producing the most interpretable, sparse coefficient set, making it the recommended model among those evaluated; further gains would likely require non-linear models (e.g. tree ensembles) given the persistent bias observed across all linear variants.

15 References

[1] Scikit-learn: Linear Models, https://scikit-learn.org/stable/modules/linear_model.html.

Scikit-learn: Hyperparameter Optimization, https://scikit-learn.org/stable/modules/grid_search.html.

Predict Loan Amount Dataset, Kaggle.